

Sweet tooth

Pierre Fauchard, a French doctor, was the first person to link the eating of sugar to tooth decay. In his book *The Surgeon Dentist*, he urged people to give up eating sugar.



Illustration from
The Surgeon Dentist,
published in 1728

1729

Electrifying experiments

Stephen Gray, a self-taught English scientist, was an early pioneer of electricity, which he produced by friction. He was able to conduct an electric charge hundreds of yards along a thread, which was draped through the house and out into the garden. He later put on public displays of electricity, including a spectacle called the Flying Boy.

1721

Smallpox protection

Variolation was a way of protecting healthy people from getting smallpox by scratching them with infected material from a smallpox scab. It became fashionable after members of the British royal family underwent the procedure.



Instrument used
to puncture the
skin in variolation

2. The charged
rod is passed over
the boy's body
to give him an
electrostatic charge.

1. A glass rod
is rubbed to
generate static
electricity.

3. Silk
cords
act as
insulators.

4. The static charge
from the boy's face
and hands attracts
pieces of paper.

5. A small ivory ball
is also attracted.

Gray's Flying Boy

1725

1725

Speedy weaving

Basile Bouchon, a French silk maker, invented the first semiautomated weaving machine. He came up with a way to speed up weaving by using a perforated paper tape to control the raising of the warp threads on the loom. His invention was the forerunner of other programmable machines, such as the computer.

Samrat Yantra
sundial

1727

Indian observatory

Jai Singh II, Maharaja of the kingdom of Jaipur, India, began constructing the Jantar Mantar astronomical observatory at Jaipur. It contained a collection of massive astronomical instruments built of brick and stone—including the world's largest sundial, Samrat Yantra.

Telling
the time
See pages
80–81

This triangular wall is
the gnomon (the part of
a sundial that casts a
shadow from the Sun).
The shadow travels at
a rate of about 0.04 in
(1 mm) per second.

This wall is a scale that
registers the position
of the shadow cast by
the gnomon.



**Samrat Yantra, the
sundial at Jantar
Mantar, has a height
of 88 ft (27 m).**

